

Analysis of LOS

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PUBLISHED  
October 5, 2026

Analysis of Traffic LOS by Facility Type

Steps

- 1. Find flow ( $v_p$ ) of the traffic stream
- 2. Find speed ( $S$ ) of the traffic Stream
- 3. Find density of traffic stream ( $D$ ) using  $v_p = D \times s$
- 4. Check appropriate curve for  $LOS$

1. Find flow ( $v_p$ ) of the traffic stream

- $$v_p = \frac{V}{PHF \times f_{HV} \times N}$$
  - $v_p$  = flow rate per lane under equivalent ideal condition (pc/hr/ln)
  - $V$  = demand volume (veh/hr)
  - $PHF$  = peak-hour factor
  - $f_{HV}$  = adj. factor for presence of heavy vehicles (HV)
  - $N$  = number of lanes in one direction

1.2 Adj. factor for presence of heavy vehicles (HV)

- $f_{HV} = \frac{1}{1 + P_{HV}(E_{HV} - 1)}$ 
  - $f_{HV}$  = adj. factor for presence of heavy vehicles (HV)
  - $P_{HV}$  = proportion of HV in traffic stream
  - $E_{HV}$  = heavy vehicle equivalent
    - $E_{HV} \sim f(\textit{Type of Terrain, Grade})$

1.2.1 Passenger Car Equivalents ( $E_{HV}$ )

- $V_{pc} = V \times P_{HV} \times E_{HV}$ 
  - $V_{pc}$  = volume in traffic stream in (pc/hr)
  - $V$  = volume in traffic stream (veh/hr)
  - $P_{HV}$  = proportion of HV in traffic stream
  - $E_{HV}$  = heavy vehicle equivalent

1.2.2  $E_{HV}$  for Terrain and Grades

- General Terrain Segments: Table 28.17
  - Condition:
    - NO one grade  $\geq 3$  longer than 0.25 *mile*
    - and NO one grad  $< 3$  longer than 0.50 *mile*
  - Level terrain
  - Rolling terrain
- Specific Grades:
  - Typical Mix of HV: Table 28.18
  - Atypical Mix of HV: Table 28.19
  - Composite Grade:
    - To find equivalent grade for a series of upgrades and/or downgrades of varying steepness.
    - Average Grade,  $G_{AV} = (\frac{\textit{total rise}}{\textit{total length}}) \times 100$
    - Use  $G_{AV}$  in Tables 28.18 and 28.19

2. Find speed ( $S$ ) of the traffic Stream

| $v_p \leq BP$   | $v_p > BP$                                                                               |
|-----------------|------------------------------------------------------------------------------------------|
| $S = FFS_{adj}$ | $S = FFS_{adj} - [\frac{(FFS_{adj} - \frac{c_{adj}}{4})(v_p - BP)^a}{(c_{adj} - BP)^a}]$ |

where,

- $FFS_{adj}$  = adjustment free-flow speed (mph)
- $BP$  = Breakpoint flow rate (pc/h/ln)
- $c_{adj}$  = adjusted capacity (pc/h/ln)
- a = calibration constant

2.1 Adjusted Free-Flow Speed  $FFS_{adj}$

| Freeways                     | Multilane Highway |
|------------------------------|-------------------|
| $FFS_{adj} = FFS \times SAF$ | $FFS_{adj} = FFS$ |

where

- $FFS_{adj}$  = Adjusted Free-Flow Speed
  - $FFS$  = Free-Flow Speed
  - $SAF$  = Speed Adjustment factor

2.1.1 Free-Flow Speed  $FFS$

| Freeways                                                | Multilane Highway                          |
|---------------------------------------------------------|--------------------------------------------|
| $FFS = 75.4 - f_{LW} - f_{LC} - 3.22 \times TRD^{0.84}$ | $FFS = BFFS - f_{LW} - f_{LC} - f_M - f_A$ |

Where

- $FFS$  = free-flow spee of the freeway (mph)
- $f_{LW}$  = adjustment for lane width (mph)

- $f_{LC}$  = adjustment for right-side lateral clearance (mph)
- $TRD$  = total ramp density (ramps / mile)
- $f_M$  = adjustment for type of median (mph)
- $f_A$  = adjustment for access points (mpg)

| Factor   | Freeways Table | Multilane Hwy Table |
|----------|----------------|---------------------|
| $f_{LW}$ | Table 28.3     | Table 28.3          |
| $f_{LC}$ | Table 28.4     | Table 28.5          |
| $f_M$    | -              | Table 28.6          |
| $f_A$    | -              | Table 28.7          |

2.1.2 Speed Adjustment Factor  $SAF$

$SAF = SAF_w \times SAF_{DP} \times SAF_{wz}$

where

- $SAF$  = Speed adjustment factor
  - $SAF_W$  = SAF for weather
  - $SAF_{DP}$  = SAF for non-std driver poulation
  - $SAF_{WZ}$  = SAF for work zone
    - $SAF_{WZ} = \frac{FFS_{WZ}}{FFS}$

| Factor     | Table       |
|------------|-------------|
| $SAF_W$    | Table 28.11 |
| $SAF_{DP}$ | Table 28.13 |
| $SAF_{WZ}$ | -           |

- $SAF_{WZ} = \frac{FFS_{WZ}}{FFS}$ 
  - $FFS_{WZ} = 9.95 + (33.49 \times f_{SR}) + (0.53 \times SL_{WZ}) - (5.60 \times LCSI) - (3.84 \times f_{BR}) - (1.71 \times f_{DN}) - (8.7 \times TRD)$ 
    - $f_{SR}$  - speed limit ratio =  $\frac{SL}{SL_{WZ}}$ 
      - $SL$  = base speed limit of the freeway (mph)
      - $SL_{WZ}$  = speed limit at work zone (mph)
    - $LCSI$  = Lane closure severity index
      - Table 28.14
    - $f_{BR}$  = adj for barrier type;
      - 0: Concrete or other hard barrier
      - 1: cones or soft barrier
    - $f_{DN}$  = adj for day or night
      - 0: day
      - 1: night
    - $TRD$  = total ramp density (ramps / mi)

2.2 Breakpoint value,  $BP$

- $BP$  Values for facility type

| Freeways                                        | Multilane Highway    |
|-------------------------------------------------|----------------------|
| $BP = [1000 + 40(75 - FFS_{adj})] \times CAF^2$ | $BP = 1,400 pc/h/ln$ |

Where

- $BP$  = Breakpoint flow
  - $FFS_{adj}$  = adjustment free-flow speed (mph)
  - $CAF$  = Capcity adjustment factor

2.1.1 Adjustment free-flow speed,  $FFS_{adj}$

- Same as section 1.1 ### 2.1.2 Capacity Adjustment Factor  $CAF$

$CAF = CAF_w \times CAF_I \times CAF_{DP} \times CAF_{wz}$

where

- $CAF$  = Speed adjustment factor
  - $CAF_W$  = CAF for weather
  - $CAF_I$  = CAF for incidents
  - $CAF_{DP}$  = CAF for non-std driver poulation
  - $CAF_{wz}$  = CAF for work zone
    - $CAF_{wz} = \frac{c_{wz}}{c}$

| Factor     | Table       |
|------------|-------------|
| $CAF_W$    | Table 28.10 |
| $CAF_I$    | Table 28.12 |
| $CAF_{DP}$ | Table 28.13 |
| $CAF_{wz}$ | -           |

- $CAF_{wz} = \frac{c_{wz}}{c}$
- Capacity of work zone,  $c_{WZ} = QDR_{WZ} \frac{100}{(100 - \alpha_{WZ})}$ 
  - $QDR_{WZ}$  = queue discharge rate from work zone
  - $QDR_{WZ} = 2,093 - (154 \times LCSI) - (194 \times f_{BR}) - (179 \times f_{AT}) + (9 \times f_{LAT}) - (59 \times f_{DN})$ 
    - $LCSI$  = Lane closure severity index
      - Table 28.14
    - $f_{BR}$  = adj for barrier type;
      - 0: Concrete or other hard barrier
      - 1: cones or soft barrier
    - $f_{AT}$  = adj for area type
      - 0: Urban
      - 1: Rural
    - $f_{LAT}$  = adj for lateral clearance to barrier

- (0 - 12)ft
- $f_{DN}$  = adj for day or night
  - 0: day
  - 1: night
- $\alpha_{WZ}$  = % reduction from capacity to QDR where

2.3 Adjusted Capacity,  $c_{adj}$

| Freeways                      | Multilane Highway             |
|-------------------------------|-------------------------------|
| $c_{adj} = c \times CAF$      | $c_{adj} = c$                 |
| $c = 2200 + 10(FFS - 50)$     | $c = 1900 + 20(FFS - 45)$     |
| $c \leq 2,400\text{ pc/h/ln}$ | $c \leq 2,300\text{ pc/h/ln}$ |
| and                           | and                           |
| $55 \leq FFS \leq 75$         | $45 \leq FFS \leq 70$         |

2.4 Calibration Constant,  $a$

| Freeways   | Multilane Highway |
|------------|-------------------|
| $a = 2.00$ | $a = 1.31$        |

3. Find density of traffic stream ( $D$ )

$$v_p = D \times s$$

4. Check appropriate curve for  $LOS$

- For Freeways: Figure 28.6
- For Multilane Highways: Figure 28.7

## Chapter 28 (Sample Problems)

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### Problem 28-4 and Problem 28-8:

A segment of an old freeway in a large urban area has the following characteristics:

- Four lanes (two lanes in each direction)
- 10 ft lane widths
- Lateral obstructions at the roadside 0 feet from the pavement edge
- Total ramp 5 ramps per mile
- Rolling terrain
- Trucks (Typical Mix) = 2%
- Peak-hour factor (PHF) = 0.95

The current peak-hour demand in the peak direction is 3000 veh/h. The roadway facility is located in an area where moderate rainfall is a common occurrence. At what level of service is the freeway expected to operate during peak periods?

## Chapter 28 (Sample Problems)

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Problem 28-5: A four-lane multilane highway segment with a full median carries a peak-hour volume of 2,600 veh/h in the heaviest direction. It has the following additional characteristics:

- 12 ft lanes
- 4 ft clearance at the roadside and in the median
- 10 access points/mile
- 10% trucks, standard mix
- Peak-hour factor = 0.88

The segment in question is on a sustained 3.5% grade, 1.25 miles in length. The base FFS may be taken as 65 mph. What is the expected level of service on the upgrade, and on the downgrade?